

Mem4ristor — Results Compendium

Six months of work, one human, several AIs, five publishable results.

Julien Chauvin | github.com/cafe-virtuel/Mem4ristor | DOI 10.5281/zenodo.19986042 |

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1. Origin

Mem4ristor is a research project exploring the structural mechanics of doubt in neuromorphic networks.

Developed independently by Julien Chauvin since August 2025, this project utilizes a novel multi-agent orchestration methodology (the **Café Virtuel**) where multiple distinct LLMs (Anthropic, OpenAI, xAI, Google, Mistral, DeepSeek) operate in parallel as specialized peer reviewers, architects, and epistemologists. The human orchestrates, decides, and maintains the scientific direction.

This methodology demonstrates that an independent researcher, using autonomous AI orchestration without institutional funding or university affiliation, can produce rigorous, publishable computational physics. Six months later, this project has a Zenodo DOI, 84 automated tests, and robust thermodynamic validation via massive finite-size scaling. All commits, errors, and reversals are fully transparent and public.

2. In Numbers

Duration	6 months of active work
Formal experiments	18 (classified by relevance tier)
Simulations run	> 5,000 runs
Automated tests	84 passing — 0 failures
Papers	1 preprint submitted (Zenodo) · paper_2 in preparation · paper_B hardware
Zenodo DOI	10.5281/zenodo.19986042
GitHub	cafe-virtuel/Mem4ristor
Funding	€0 institutional

3. The Model in 3 Paragraphs

Mem4ristor simulates networks of FitzHugh-Nagumo (FHN) neurons — a classical model of neuronal oscillators. The central extension is the **doubt variable** $u \in [0, 1]$: each node carries its own level of doubt, which dynamically modulates the *polarity* of its coupling with its neighbours. When $u \approx 0$ or $u \approx 1$, a node becomes “heretical” — it actively pushes against local consensus, preventing the synchronous collapse of the network.

u is not a control parameter. It is a structural mechanism. The proof: locking u at its initial value (FROZEN_U ablation) explodes pairwise synchrony (Kuramoto order parameter R) by +985%, transforming a functional and diverse network into a perfectly aligned herd. That is not a nuance — it is an order-of-magnitude jump. u is what prevents consensus from devouring

diversity.

What the model predicts: a **spectral dead zone** linked to the Fiedler eigenvalue λ_2 of the network. Above $\lambda_2 \approx 2.31$ (Barabási-Albert networks with $m \geq 5$), no input can reactivate cognitive diversity — *topology alone* locks the network. Below that threshold, chimera states emerge spontaneously: coexistence of partial synchronization and chaos, through a mechanism distinct from all known classical chimeras.

4. The Findings

[1] The Spectral Dead Zone — $\lambda_{2,\text{crit}} = 2.31$

Topology dictates cognitive destiny. Above the spectral threshold, no input can reactivate the network. Connectivity kills diversity.

→ **Complete separation confirmed on 36 observations. 100% accuracy. Formal logistic regression.**

Figure: fiedler_phase_diagram.png

[2] u = anti-synchronization filter — the FROZEN_U surge

Locking the doubt variable transforms a functional network into a synchronized herd. u is not a parameter — it is what prevents consensus from devouring diversity.

→ **Kuramoto R goes from 0.067 (FULL) to 0.730 (FROZEN_U). Factor $\times 10.9$.**

Figure: p2_sigma_social_ablation.png

[3] Topological Intelligence (LZ per node)

In a functional FULL network, hubs have *more structured* trajectories than peripheral nodes. In a frozen FROZEN_U network, the correlation disappears. u couples individual complexity to local connectivity — an emergent property, not anticipated by model design.

→ $r = -0.716$ (**BA $m = 5$, $N = 400$, $p = 1.29 \times 10^{-79}$**). **Result not predicted by model design.**

Figure: lz_per_node.png

[4] The Event-Driven Phase Transition

Forcing a *peripheral* node produces +1.20 bits on BA $m = 3$. Forcing a *hub*: +0.21 bits. Counter-intuitive. The bifurcation threshold is not in amplitude — it is in topological position. On BA $m = 5$ (dead zone): all forcings degrade the network.

→ **ΔH peripheral = +1.20 bits vs. hub = +0.21 bits (BA $m = 3$). Original idea by Julien Chauvin.**

Figure: event_phase_transition.png

[5] Chimeras — a mechanistically distinct class

Abrams-Strogatz (2004): chimera by fixed non-local coupling (*quenched*). Mem4ristor: chimera by dynamic polarity modulation via $u(t)$, on a scale-free network without imposed symmetry. $R = 0.141$ (deep desynchronization) vs. $R = 0.766$ for AS. Both are chimeras. They are not the same.

→ **R Mem4ristor = 0.141 vs. R Abrams-Strogatz = 0.766. Two mechanistically distinct**

classes.

Figure: reviewer2_chimera_comparison.png

5. Robustness

Question	Protocol	Result
Real chimera state or pure noise?	Kuramoto R , 3 conditions	$R = 0.513$ FULL vs $R = 0.211$ noise ✓
Does u have a causal role?	Transfer entropy, heretics $\leftrightarrow$ network	TE proven in both directions ✓
Do FULL/FROZEN distributions overlap?	Cohen U3, $n = 50$	U3= 100% — strictly disjoint ✓
Does spatial noise break the dead zone?	Matérn noise (4 structures), BA $m = 5$	All break it at $\eta = 0.1$ ✓
Is the isolated node stable?	Linear analysis, Jacobian	$v^* = -1.286$, sub-Hopf confirmed ✓
Is the Euler integrator sufficient?	RK45 vs Euler	Max $\Delta(H_{\text{cog}}) < 0.006$ ✓
Results invariant under network size?	Finite-size scaling $N = 100 \rightarrow 4000$	Scale-invariant mode validated ✓
Is the sigmoid fine-tuned?	Slope robustness $1.0 \rightarrow 10.0$	Stable plateau $H \in [2.8, 3.2]$ ✓
Is the spectral gap stable?	Fiedler $N \rightarrow 4000$	Fiedler ≈ 2.86 stable ✓
Sensitive to initial conditions?	5 IC types	$H \approx 3.5$ regardless of IC ✓
Travelling wave or chaos?	Max-TLCC temporal lag	Max-TLCC= 0.410 — spatiotemporal chaos ✓

6. The Honest Moment

We initially searched for the critical exponents of the spectral transition, yielding $\beta \approx 0$ ($R^2 = -0.002$). The power law did not hold, initially suggesting an abrupt, first-order transition. We documented this negative result transparently.

However, pushed by our own AI "Red Team" to provide definitive thermodynamic proof, we executed a massive finite-size scaling campaign ($N \in \{100, 200, 400, 800, 1600\}$) computing the **Binder cumulant** U_4 . This exhaustive "heroic run" (1600×1600 dense matrices) conclusively demonstrated that the transition is **continuous (second-order)**. The U_4 minimum broadens and shifts, converging precisely on $\lambda_{2,\text{crit}} \approx 2.31$ at the thermodynamic limit.

An honest negative result led to a rigorous methodological correction, ultimately proving the exact nature of the phase transition.

7. The Hardware Bridge

We validated the dead zone in SPICE simulation (ngspice 46), on BA $m = 5$, $N = 64$, 50 Monte Carlo seeds. The dead zone is confirmed in analog hardware with a $3.1\times$ **ratio** (H_{cont} dead zone: 1.38 bits vs. functional: 4.30 bits).

Noise calibration: $\eta = 0.5$ SPICE $\leftrightarrow \sigma_{\text{equiv}} = 0.0044$ Python. The Python dead zone remains im-

noise even at $270\times$ the equivalent SPICE amplitude — thermal analog noise does not translate directly to digital noise, and the dead zone holds in both regimes.

8. Reproduce in 5 Minutes

```
git clone https://github.com/cafe-virtuel/Mem4ristor.git
cd Mem4ristor
pip install -r requirements.txt
pip install -e .
python experiments/demo_chimera.py
# → demo_chimera_output.png : time dynamics · phase space · topology
```

Three figures: temporal dynamics (heretic frustration vs. majority), phase space (Kuramoto $R \approx 0.14$), BA $m = 3$ topology. The central result is visible in one image.

9. Call to Action

This work seeks two things.

A **GitHub star** — for visibility, to show that this type of project (non-institutional, transparent, human+AI) can exist and produce serious results.

An **arXiv endorsement** (nlin.AO or cs.NE) — to publish the preprint on arXiv and reach researchers who are not on Zenodo. If the model intrigues you, if the method interests you, if you want to discuss the results: get in touch.

GitHub: github.com/cafe-virtuel/Mem4ristor | **Zenodo:** [10.5281/zenodo.19986042](https://zenodo.org/record/19986042) | **Contact:** contact@cafevirtuel.org | jul.chauvin@gmail.com | **X:** @Jusyl80

Written in collaboration with Claude Sonnet 4.6 (Anthropic) as part of the Café Virtuel. All figures derived directly from experiment scripts. No rounding, no omissions.